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Abstract

Glaucoma, a leading cause of irreversible blindness, often progresses silently until significant vision loss occurs. Early diagnosis is crucial for effective management, but conventional screening methods like visual field testing and optic disc assessment are subjective, labour-intensive, and prone to variability among clinicians.

Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs), have shown remarkable potential in automated glaucoma detection and prediction from retinal fundus images. CNN-based models can capture subtle structural changes in the optic nerve head and retinal nerve fibre layer, even years before clinical manifestation. Studies demonstrate that CNNs, when trained on large-scale fundus datasets, achieve high diagnostic accuracy and area under the curve (AUC), enabling robust classification of glaucomatous and non-glaucomatous eyes.

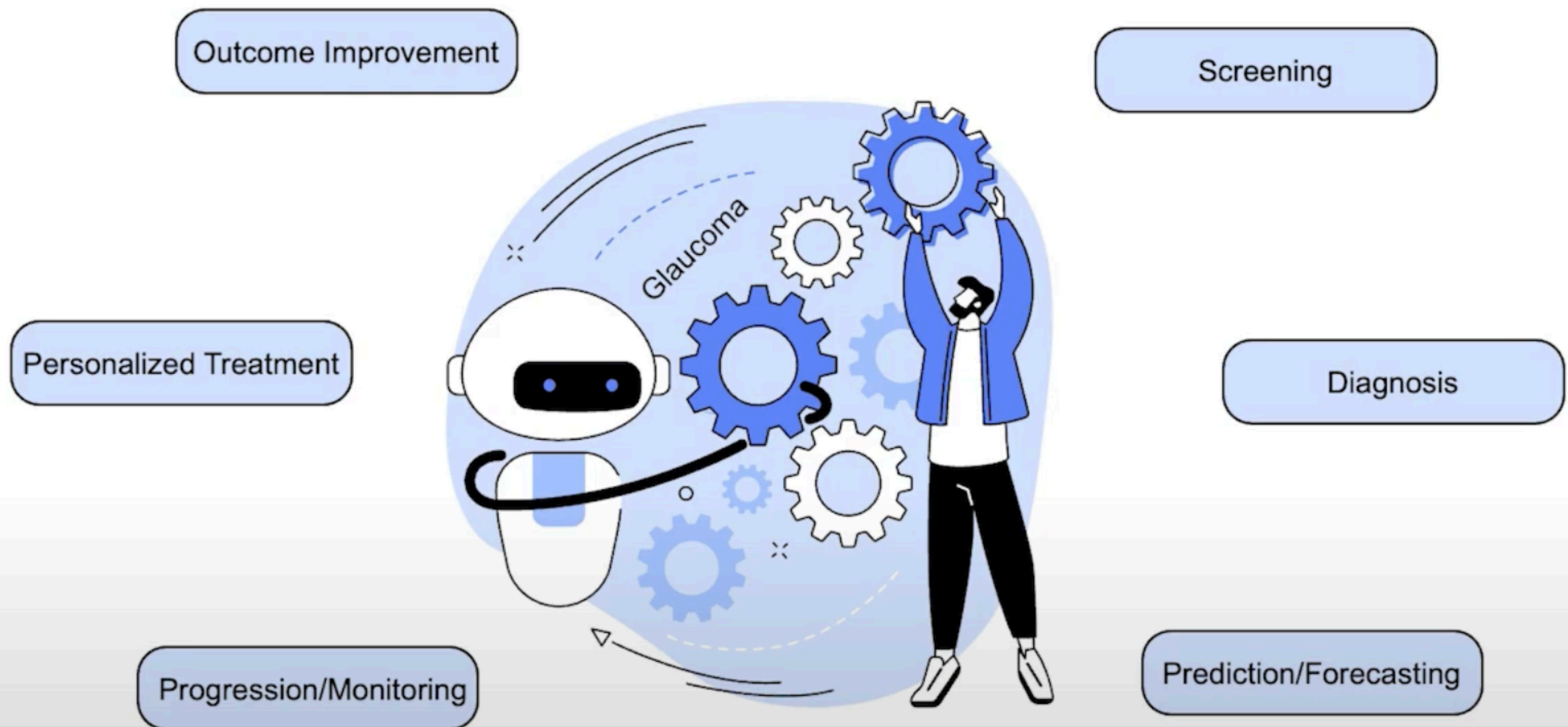
These automated systems reduce clinical workload, offer scalable, cost-effective, and objective tools for large-scale screening and early intervention, ultimately improving patient outcomes.

Keywords: Glaucoma, Convolutional Neural Networks (CNN), Deep Learning, Computer Vision, Retinal Fundus Images, Automated Screening, Medical Image Analysis.

Introduction

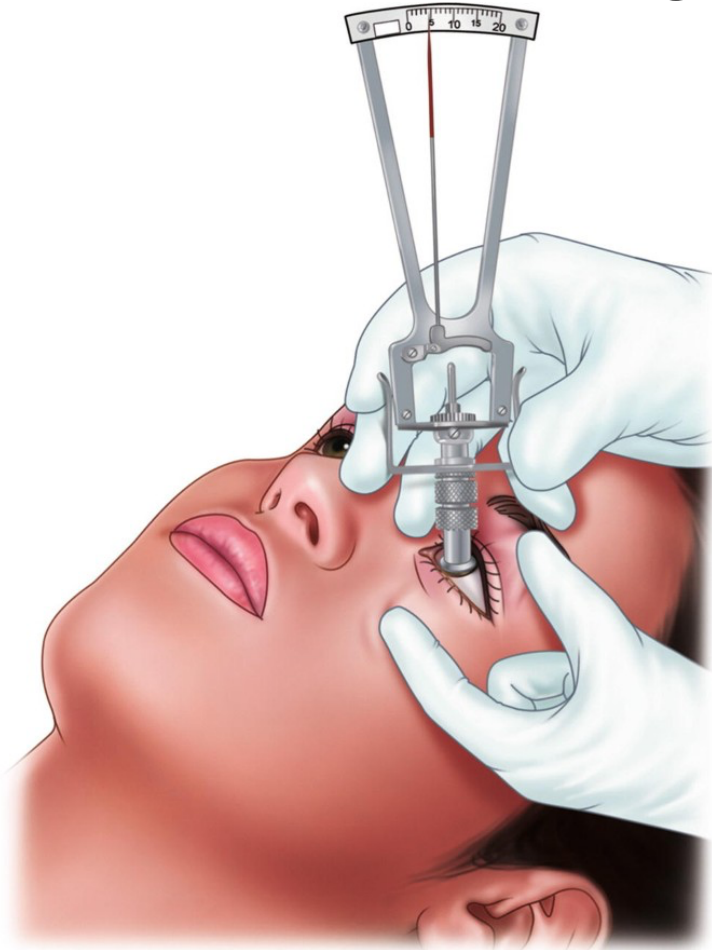
- Glaucoma is a Eye disease that damages the optic nerve due to increased intraocular pressure .
- It is one of the cause irreversible blindness.
- Early detection of glaucoma can prevent the vision loss.
- It's not easy to detect glaucoma, at early stages, as there are no noticeable symptoms in its early stages, earning it, the nickname, "the silent thief of the sight".

Artificial Intelligence in Glaucoma



Literature survey

- ▶ Initially tonometry is used to measure intraocular pressure.
- ▶ Raise in Intraocular pressure is one of the risk factor of glaucoma but, sometimes normal IOP can also lead to Glaucoma.
- ▶ So, we need to take use of technologies like Artificial Intelligence and other technologies to predict and screen glaucoma at early stages.

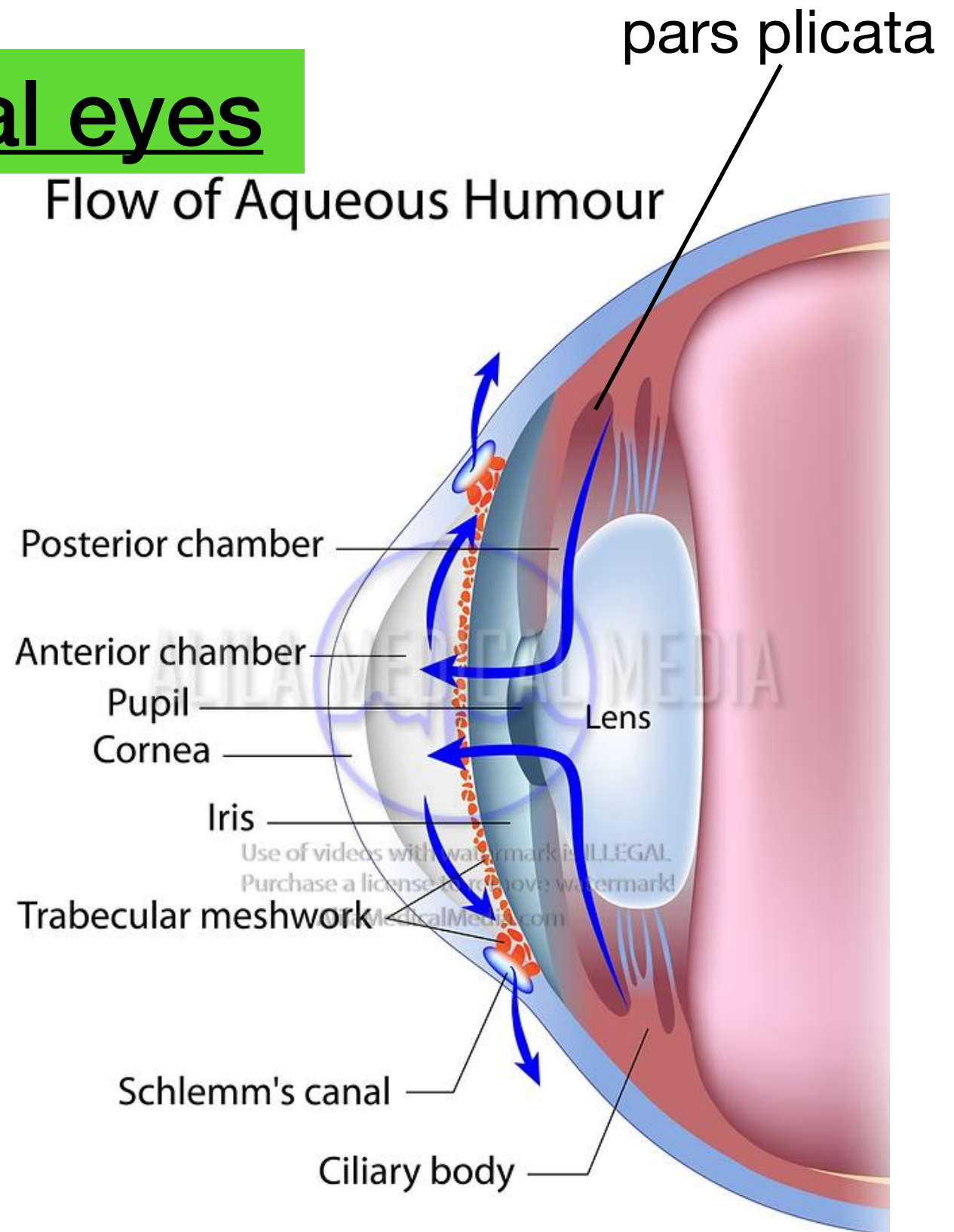


Problem statement

- Glaucoma risk factors include age, family history, ethnicity, high intraocular pressure, certain medical conditions, eye injuries, and prolonged corticosteroid use.
- It must be detected early. To cure the disease it is most important to find and start its treatment as early as possible.
- Sometimes it is very easy for an Ophthalmologists to oversee glaucoma in the eye. A trained eye can also make mistakes while detecting glaucoma and they can overlook it.
- This is where we need to take help from technology and solve this problem by using technology on it.

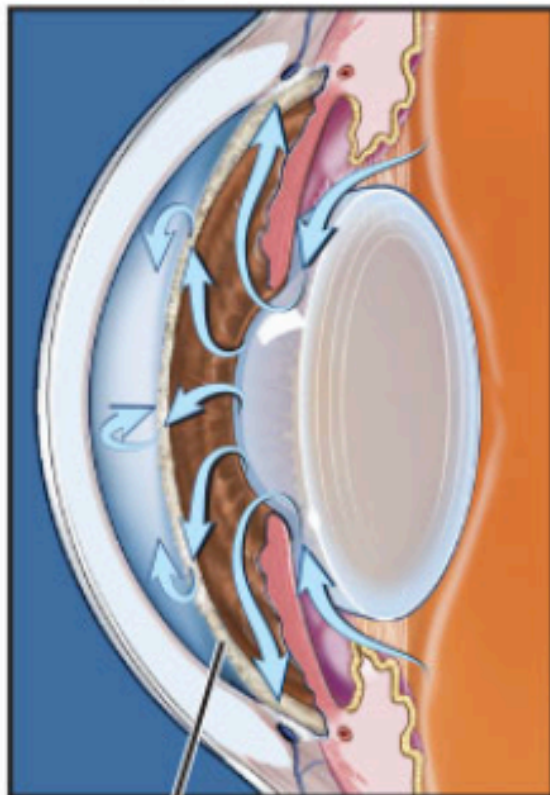
Normal eyes

- Aqueous humour is produced at the pars plicata and travels from the posterior chamber to the anterior chamber.
- Approximately 90% of aqueous humour drains through Schlemm's canal, where it mixes with retinal veins.
- Schlemm's canal may not be able to release aqueous humour due to various reasons, such as prolonged use of corticosteroids, genetic factors, and other conditions.
- This condition is known as "**Glaucoma**".



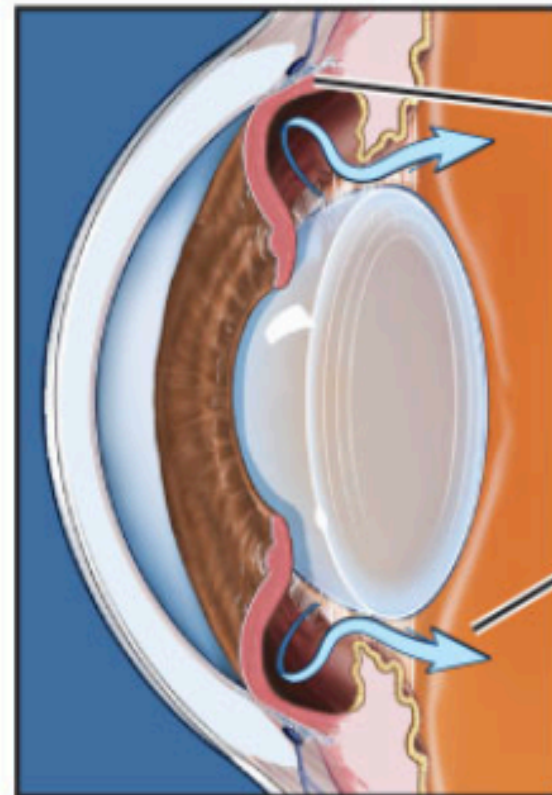
Glaucoma

Open-angle glaucoma



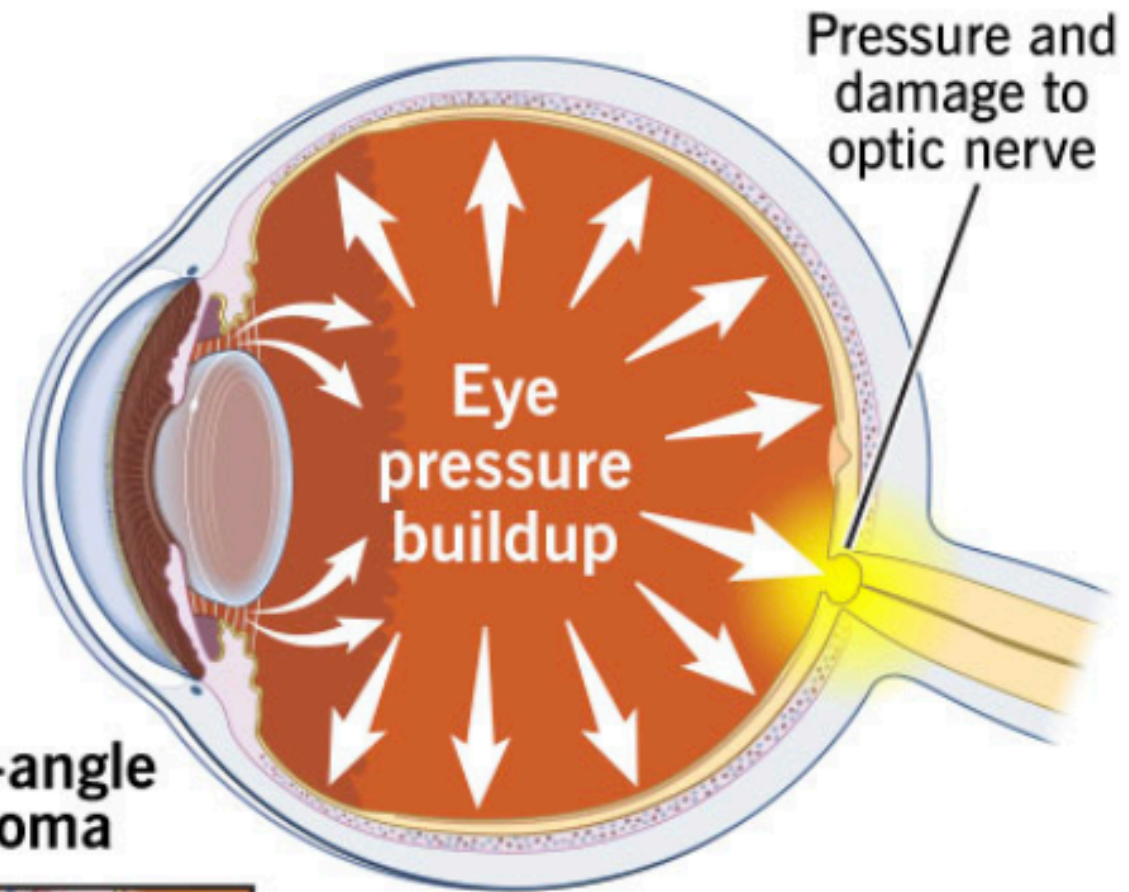
Drainage canals are clogged, fluid pressure builds up

Closed-angle glaucoma



Angle between the iris and cornea is too narrow

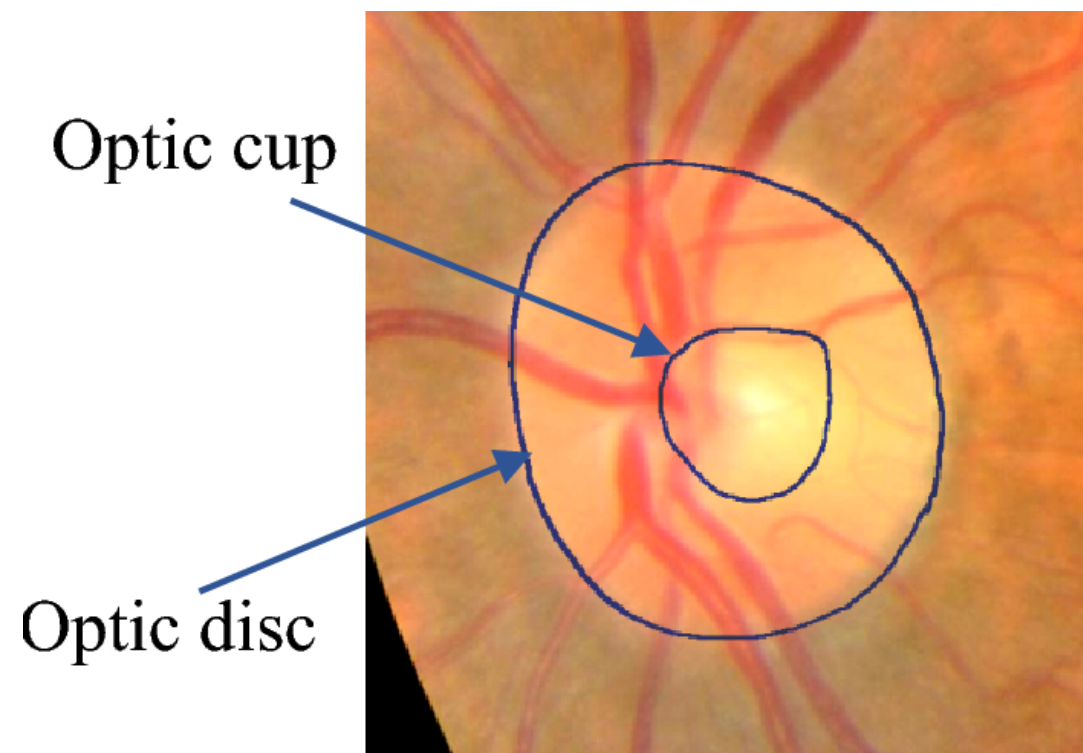
Fluid pressure builds up



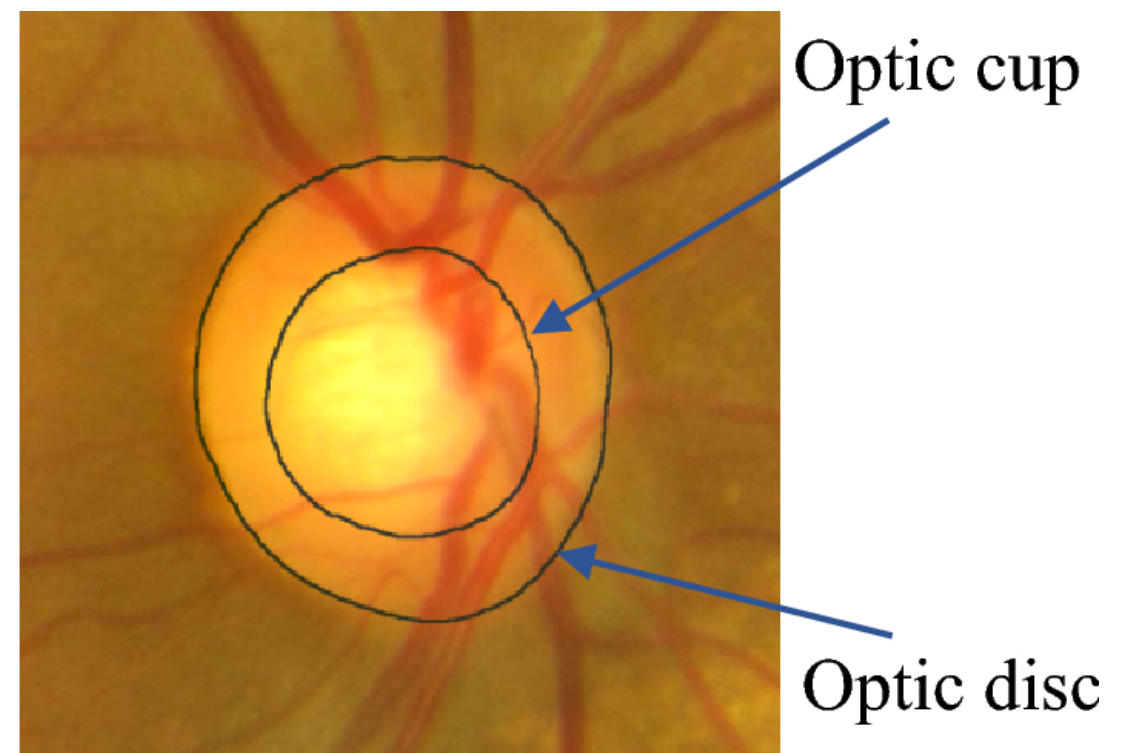
Signs of glaucoma

1) **Optic cupping** : Enlargement of optic cup (central part of optic disc).

- Normal c:d ratio is 0.3.
- In optic cupping c:d ratio is >0.5 .



Healthy

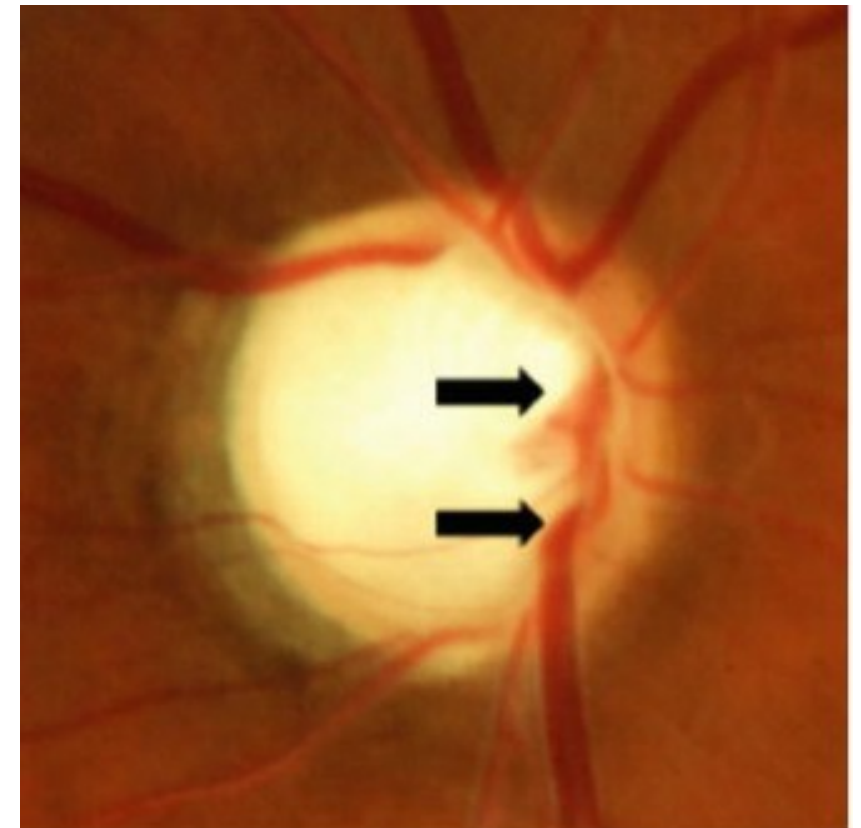


Cupped

2)Nasalisation: *The displacement of blood vessels,towards the nasal (inner) side of the optic disc.*



Healthy



Nasalised

3) Bayonetting sign: Condition where a blood vessel bends sharply at two distinct points(Double-angulation of blood vessels)



Healthy



Double-angled

Proposed methodology

- **Computer vision has become the most widely used field of Artificial intelligence. It has made its way into the medical field too.**
- **Nowadays artificial intelligence is used in medical field to diagnose a disease, optimize the treatments and many more.**
- **Deep learning algorithms are being used to make diagnostic models to detect the disease at an early stage.**

- A CNN (Convolutional Neural Network) is a type of deep learning algorithm that excels at analyzing images.
- In the context of glaucoma detection, CNNs are used to analyze retinal images, identify subtle patterns indicative of the disease, and classify images as either glaucomatous or non-glaucomatous.
- CNNs can achieve high accuracy in detecting glaucoma, even in subtle or early stages of the disease .

Applications of CNN



Computer Vision and related application



Natural Language Processing



Object Detection and Segmentation

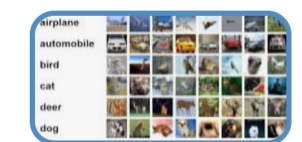
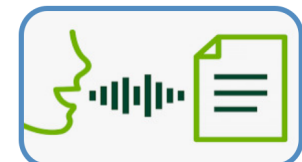
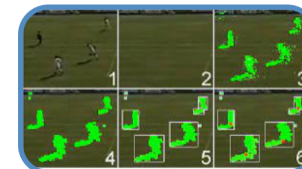


Image Classification



Speech Recognition



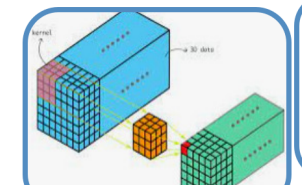
Video Processing



Low-resolution images



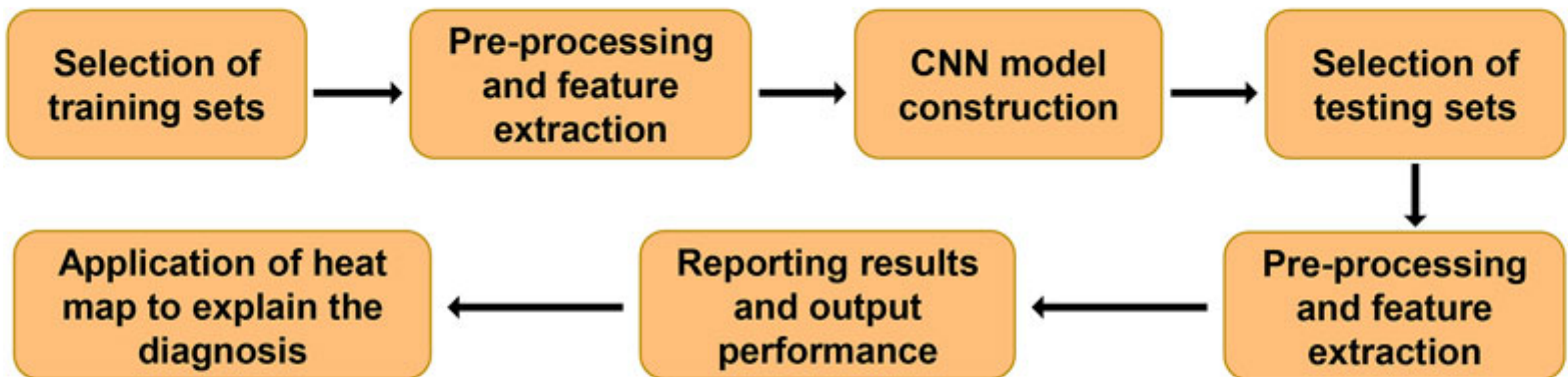
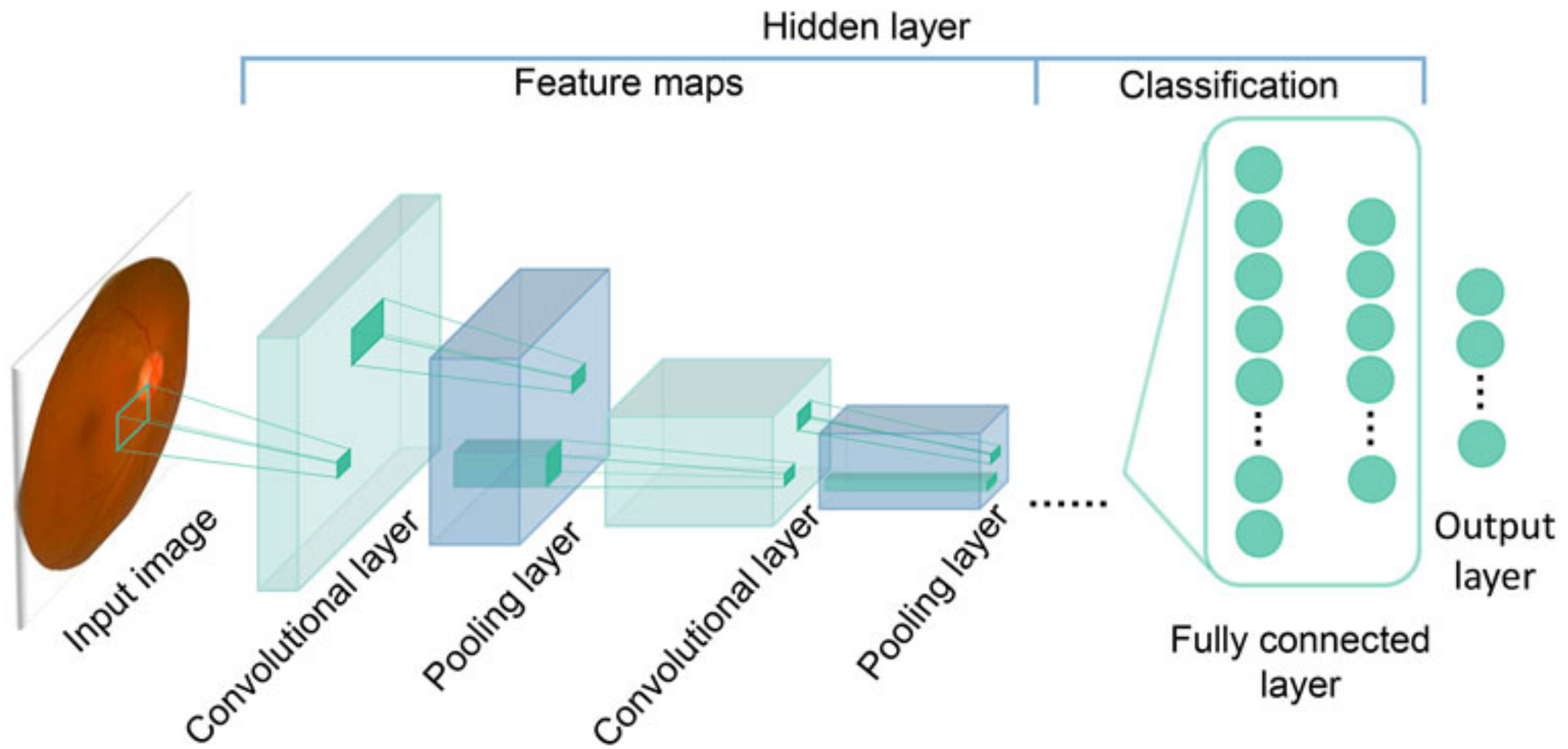
Limited Resource system



CNN for various dimensional data



Object Counting



Predicting Glaucoma before Onset Using Deep Learning

Anshul Thakur, PhD,^{1,2} Michael Goldbaum, MD, MSc,³ Siamak Yousefi, PhD^{2,4}

Purpose: To assess the accuracy of deep learning models to predict glaucoma development from fundus photographs several years before disease onset.

Design: Algorithm development for predicting glaucoma using data from a prospective longitudinal study.

Participants: A total of 66 721 fundus photographs from 3272 eyes of 1636 subjects who participated in the Ocular Hypertension Treatment Study (OHTS) were included.

Main Outcome Measures: Accuracy and area under the curve (AUC).

Methods: Fundus photographs and visual fields were carefully examined by 2 independent readers from the optic disc and visual field reading centers of the OHTS. When an abnormality was detected by the readers, the subject was recalled for retesting to confirm the abnormality and for further confirmation by an end point committee. By using 66 721 fundus photographs, deep learning models were trained and validated using 85% of the fundus photographs and further retested (validated) on the remaining (held-out) 15% of the fundus photographs.

Results: The AUC of the deep learning model in predicting glaucoma development 4 to 7 years before disease onset was 0.77 (95% confidence interval [CI], 0.75–0.79). The accuracy of the model in predicting glaucoma development approximately 1 to 3 years before disease onset was 0.88 (95% CI, 0.86–0.91). The accuracy of the model in detecting glaucoma after onset was 0.95 (95% CI, 0.94–0.96).

Conclusions: Deep learning models can predict glaucoma development before disease onset with reasonable accuracy. Eyes with visual field abnormality but not glaucomatous optic neuropathy had a higher tendency to be missed by deep learning algorithms. *Ophthalmology Glaucoma* 2020;3:262-268 © 2020 Published by Elsevier Inc. on behalf of the American Academy of Ophthalmology



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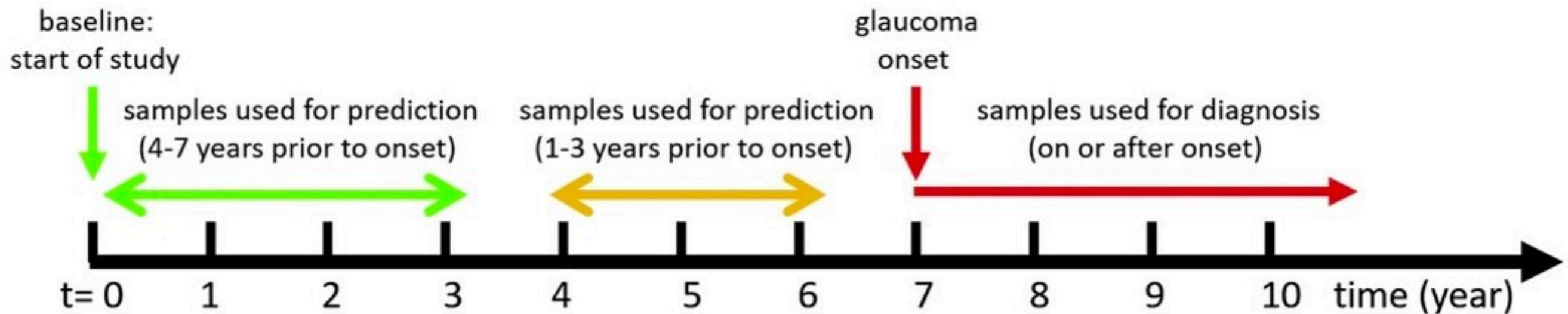
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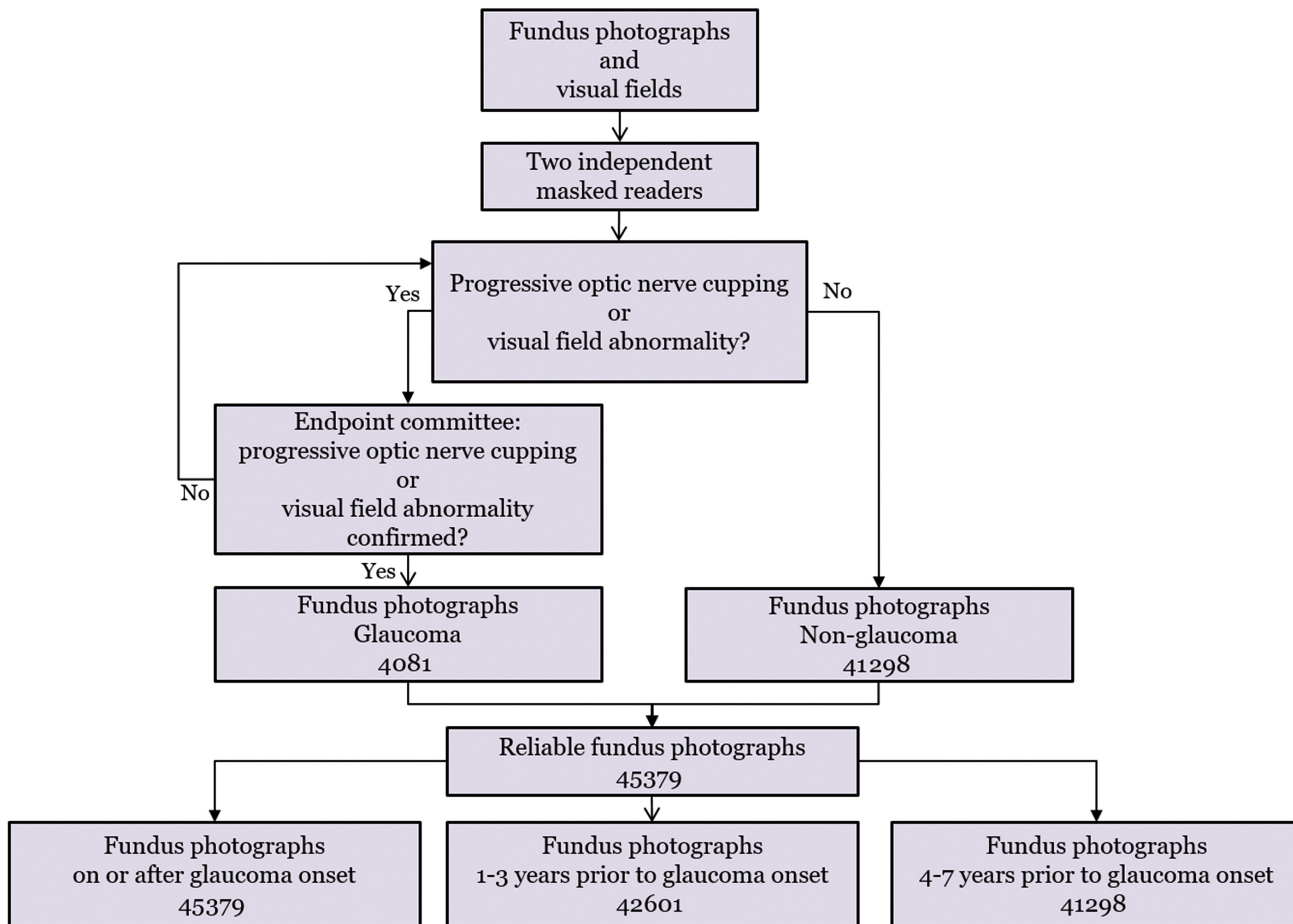
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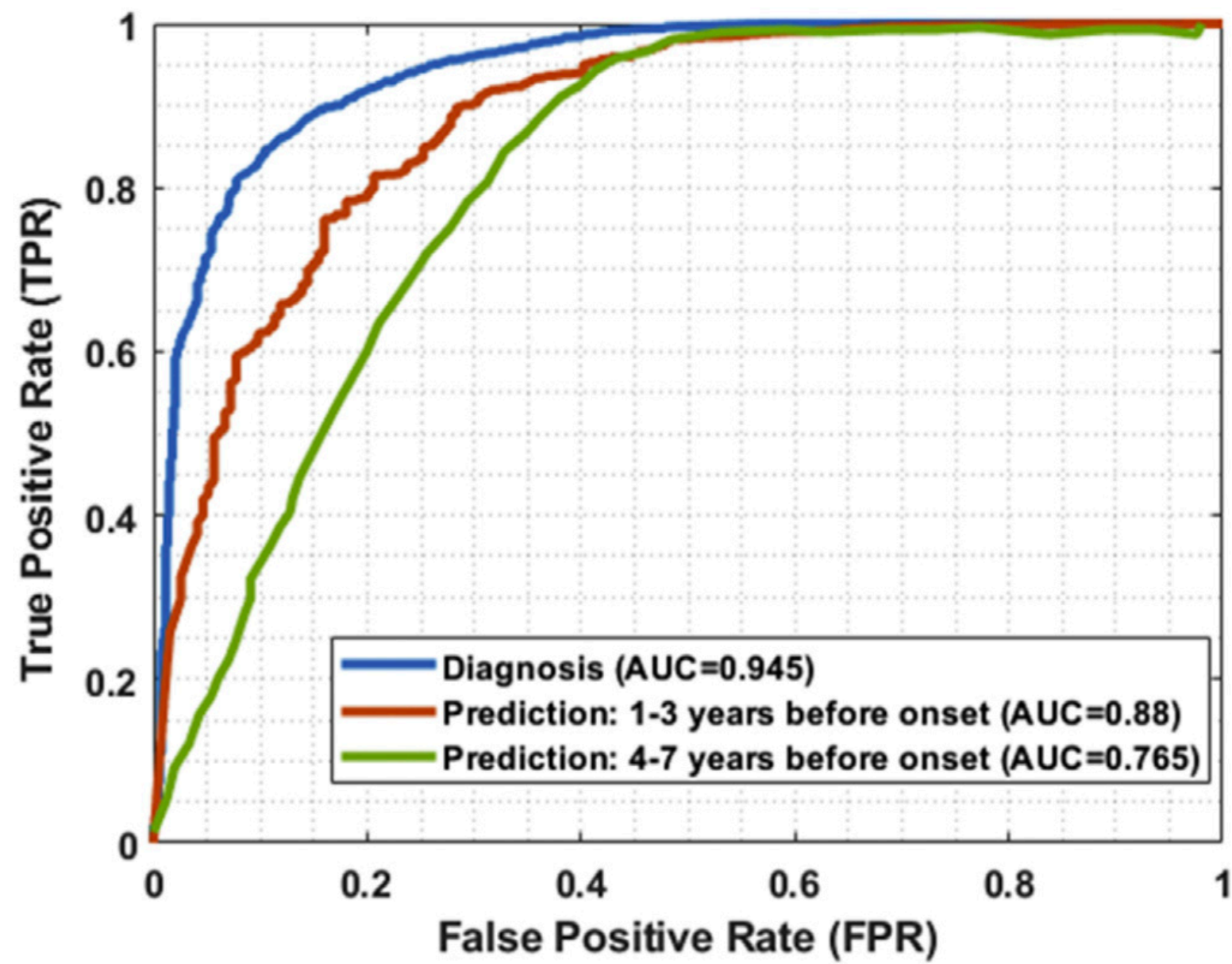
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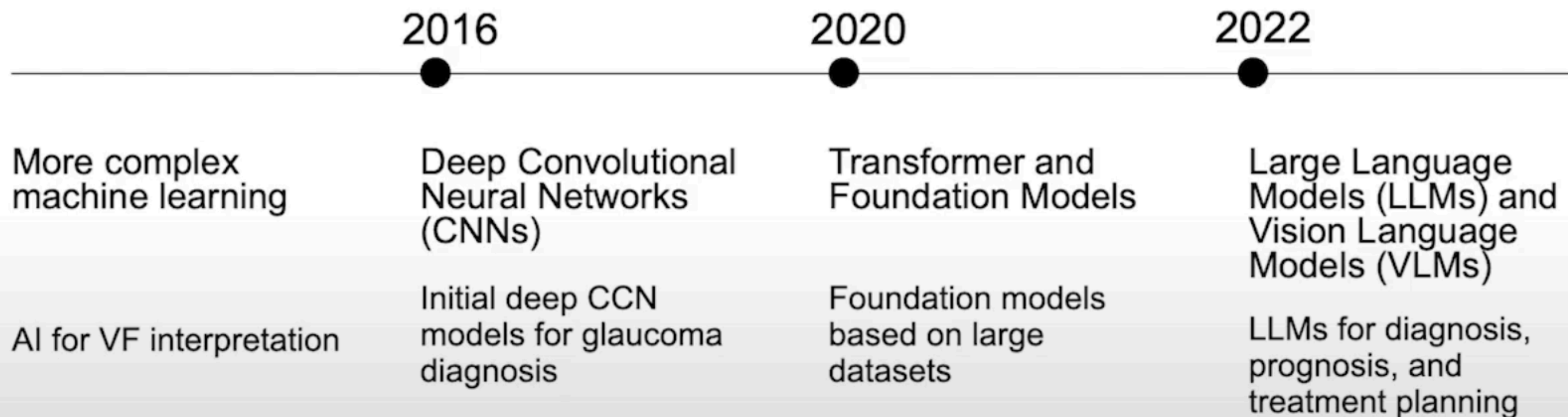
Experimental Results



Conclusion and future scope

- Early detection of glaucoma can help prevent serious vision loss.
- CNNs can analyze retinal images and identify glaucoma-related patterns.
- AI can support doctors in screening and early assessment.
- Larger and clinically validated datasets are needed for reliable real-world use.
- AI + Ophthalmology → Earlier detection and better patient care.
-

AI in ophthalmology and glaucoma



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THANK YOU

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